ALK fusion-hotspot benchmark report

Study: luad_tcga_pan_can_atlas_2018

Abstract

This report analyzes ALK gene fusions from the luad_tcga_pan_can_atlas_2018 study, covering 5 fusion events. 5/5 fusions with known frame status (100.0%) were in-frame. 5/5 fusions (100.0%) retained the Protein kinase domain. Domain retention was tested with Fisher's exact test (p=1, not statistically significant at alpha=0.05). No literature benchmark is configured for ALK.

Results summary

Fusion partner frequency

Fusion-partner frequency was tabulated across 5 analyzed fusion events, identifying 1 distinct fusion partner gene. The most frequent partner was EML4, observed in 5 events.

Domain retention

Protein kinase domain retention was tested with Fisher's exact test comparing in-frame fusions against all others; 5/5 (100.0%) of in-frame fusions retained the domain, p=1 (not statistically significant at alpha=0.05). A breakpoint-position permutation test produced a corroborating empirical p-value of 0.82018.

Domain disruption

Domain-disruption analysis was skipped: ALK has no disruption_required_domains configured; domain-disruption analysis was skipped.

Junction concentration

100% of 5 fusions with this gene as the 3' partner join at exon 20, against 0% expected if breakpoints fell at random along the gene; this is the most over-represented junction (empirical p=0.000999001, corrected for testing every exon, statistically significant at alpha=0.05). The random-breakage null assumes a breakpoint anywhere in the gene would be detected. Capture panels such as MSK-IMPACT tile only selected introns, so concentration on panel data is partly by assay design and this p-value overstates the evidence for biological concentration. Confirm in a cohort sequenced without intron selection (whole genome or RNA).

Cutpoint detection

Cutpoint detection could not determine a breakpoint boundary for this run: fewer than 2 distinct breakpoint positions to scan

Corroborating confidence statistics

Corroborating confidence statistics compared Domain_retention_flags groups "retained" (n=5) and "not_retained" (n=0). For Frame_status, retained: 5/5 (100.0%, 95% CI 56.6%-100.0%); not_retained: 0/0 (unavailable, 95% CI unavailable-unavailable). Welch's t-test comparing Tumor_variant_count between groups gave means of unavailable vs unavailable, p=unavailable.

Composite evidence score

Composite evidence score reported: domain_disruption sub-score excluded: the supplied result had no applicable p-value (e.g. disruption_required_domains not configured for this gene).

Exon retention

Exon retention produced results for this run (1 row(s) in Exon_retention); see the accompanying table.

Expression association

Expression association reported: ALK: fusion-positive-vs-negative expression comparison skipped; no sample had both established SV assay coverage and measured expression, so no fusion-negative group could be defined.

Genomic position recurrence

Genomic position recurrence produced results for this run (1 row(s) in genomic_bin_recurrence, 1 row(s) in exact_position_recurrence, 1 row(s) in protein_position_genomic_spread); see the accompanying table.

Joint partner

Joint partner reported: ALK has no gene_pair configured; joint-partner analysis was skipped.

Mechanistic interpretation

Mechanistic interpretation produced results for this run; see the accompanying data for details.

Mutation cooccurrence

Mutation cooccurrence reported: ALK has no mutual_exclusivity_targets configured; co-occurrence/mutual-exclusivity analysis was skipped.

Window detection

Window detection reported: fewer than 2 distinct breakpoint positions to scan

Tables

composite_score tables

composite_evidence_ranking

Partner_gene	Event_count	Recurrence_score	Domain_retention_score	Domain_disruption_score	Cutpoint_proximity_score	Confidence_certainty_score	Components_applicable	Composite_score	Rank
EML4	5	1.0	0.008609092035987788	None	None	0.5655175352168251	confidence_certainty, domain_retention, recurrence	0.5518523485087377	1

domain_retention tables

frame_domain_contingency_table

5	0
0	0

frame_domain_contingency_tables_by_tumor_type

stratum	table
LUAD	[5, 0], [0, 0]

domain_retention_descriptives

Domain_key	Domain_name	Quantitative_call_count	Fully_retained_count	Truncated_count	Fully_lost_count	Mean_retained_fraction_among_non_retained_calls
kinase	Protein kinase domain	5	5	0	0	None

exon_retention tables

Exon_retention

Exon	Retained_event_count	Total_event_count	Retained_fraction
20	5	5	1.0

frequency tables

Partner_gene_counts

Partner_gene	Event_count	Raw_event_count	Sample_count	Patient_count
EML4	5	5	5	5

genomic_position_recurrence tables

genomic_bin_recurrence

chromosome	bin_start	bin_end	n_events	event_ids
2	29223000	29224000	5	EVT-TCGA-50-8460-01-1, EVT-TCGA-67-6215-01-2, EVT-TCGA-67-6216-01-3, EVT-TCGA-78-7163-01-4, EVT-TCGA-86-A4P8-01-5

exact_position_recurrence

chromosome	position	n_events	event_ids
2	29223528	5	EVT-TCGA-50-8460-01-1, EVT-TCGA-67-6215-01-2, EVT-TCGA-67-6216-01-3, EVT-TCGA-78-7163-01-4, EVT-TCGA-86-A4P8-01-5

protein_position_genomic_spread

protein_position_aa	n_events	n_distinct_genomic_positions	genomic_span_bp	spans_multiple_chromosomes
1058	5	1	0	False

junction_concentration tables

junction_counts

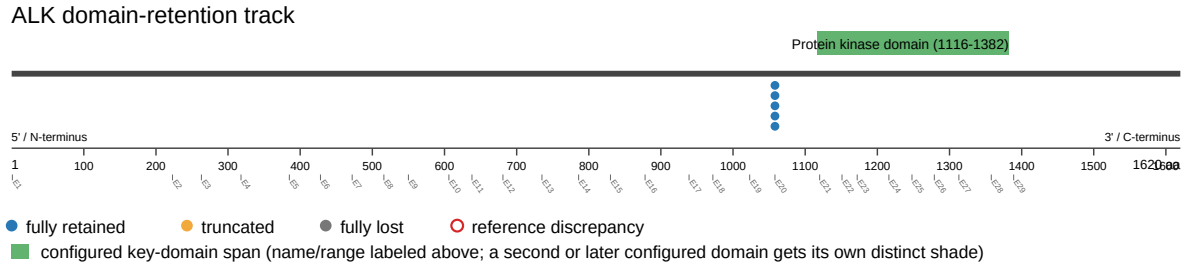
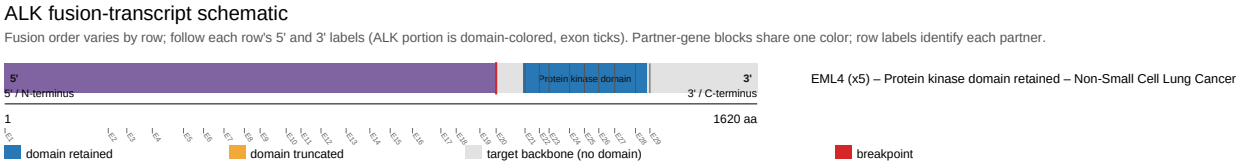
role	junction_exon	count	role_event_count	observed_share	random_breakage_share	excess_neg_log10_p
three_prime	20	5	5	1.0	0.0029075471361552593	12.68236617954754

Per-event results (results.tsv)

event_id	sample_id	patient_id	tumor_type	oncotree_code	fusion_name	partner_gene	frame_status	target_role	breakpoint_genomic	breakpoint_exon	breakpoint_protein_position	is_intronic_breakpoint	fusion_annotation_crosscheck_protein_position	fusion_annotation_crosscheck_agrees	fusion_annotation_crosscheck_detail	domain_status	domain_retained_fraction	domain_is_truncated	domain_reconstitution	reconstitution_pfam_accession	reconstitution_five_prime_span	reconstitution_three_prime_span	reconstitution_junction_gap_fraction	reconstitution_junction_gap_residues	reconstitution_partner_breakpoint_protein_position	reconstitution_partner_domain_start_aa	reconstitution_partner_domain_end_aa	reconstitution_partner_breakpoint_mid_exon	reconstitution_detail	retained_domains	lost_domains	disrupted_domains	source_annotation_text	source_site2_effect_on_frame
EVT-TCGA-50-8460-01-1	TCGA-50-8460-01	TCGA-50-8460	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29223528	20	1058	False	1058	True	exact match	retained	1.0	False												Protein kinase domain			NA | EML4-ALK fusion	in-frame
EVT-TCGA-67-6215-01-2	TCGA-67-6215-01	TCGA-67-6215	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29223528	20	1058	False	1058	True	exact match	retained	1.0	False												Protein kinase domain			NA | EML4-ALK fusion	in-frame
EVT-TCGA-67-6216-01-3	TCGA-67-6216-01	TCGA-67-6216	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29223528	20	1058	False	1058	True	exact match	retained	1.0	False												Protein kinase domain			NA | EML4-ALK fusion	in-frame
EVT-TCGA-78-7163-01-4	TCGA-78-7163-01	TCGA-78-7163	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29223528	20	1058	False	1058	True	exact match	retained	1.0	False												Protein kinase domain			NA | EML4-ALK fusion	in-frame
EVT-TCGA-86-A4P8-01-5	TCGA-86-A4P8-01	TCGA-86-A4P8	Non-Small Cell Lung Cancer	LUAD	EML4-ALK	EML4	in-frame	three_prime	29223528	20	1058	False	1058	True	exact match	retained	1.0	False												Protein kinase domain			NA | EML4-ALK fusion	in-frame

Figures

domain retention outliers



fusion schematic